

Lab 4 Submission

Question 1

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Oct 31 8:44 PM

1/1 + [ ] [ ] [ ] [ ]
Tilix: deb@cis106: ~

1: deb@cis106: ~
deb@cis106:~$ apt search "tetris clone"
bastet/stable 0.43.2-2 arm64
  ncurses Tetris clone with a bastard algorithm

ltris/stable 2.0.3-1 arm64
  very polished Tetris clone with CPU opponents

tint/stable 0.07+b1 arm64
  Tetris clone for text based terminal

deb@cis106:~$

3: deb@cis106: ~
deb@cis106:~$ apt search --names-only "tetris"
games-tetris/stable 5 all
  Debian's tetris-like games

vitetris/stable 0.59.1-3+b2 arm64
  Virtual terminal *tris clone

deb@cis106:~$

2: deb@cis106: ~
bastet/stable 0.43.2-2 arm64
  ncurses Tetris clone with a bastard algorithm

blockattack/stable 2.9.0-1+b1 arm64
  puzzle game inspired by Tetris

blockout2/stable 2.5+dfsg1-2 arm64
  Tetris like game (3D-tetris)

blocks-of-the-undead/stable 1.0-7+b1 arm64
  Tetris Attack clone with spooky undertones

blocks-of-the-undead-data/stable 1.0-7 all
  Tetris Attack clone with spooky undertones - data files

lines 1-15
This site uses JUST THE DOCS, a
documentation theme for Jekyll.
descriptions (and other searchable metadata).
```

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Oct 31 9:02 PM

1/1 + [ ] [ ] [ ] [ ]
Tilix: deb@cis106: ~

1: deb@cis106: ~
deb@cis106:~$ sudo apt install blockout2+ cuyo+ blockattack-
The following packages were automatically installed and are no longer required:
  libfmt10 libphysfs1 libSDL2-mixer-2.0-0 libxmp4
  libopusfile0 libSDL2-image-2.0-0 libSDL2-ttf-2.0-0
Use 'sudo apt autoremove' to remove them.

Installing:
  blockout2  cuyo

Installing dependencies:
  cuyo-data  libmikmod3  libSDL2-mixer1.2  libSDL1.2debian

REMOVING:
  blockattack

Summary:
  Upgrading: 0, Installing: 6, Removing: 1, Not Upgrading: 0
  Download size: 8,041 kB
  Space needed: 4,400 kB / 26.7 GB available

Continue? [Y/n] y
Get:1 http://deb.debian.org/debian trixie/main arm64 libmikmod3 arm64 3.3.13-1 [127 kB]
Get:2 http://deb.debian.org/debian trixie/main arm64 libSDL1.2debian arm64 1.2.68-3 [83.1 kB]
Get:3 http://deb.debian.org/debian trixie/main arm64 libSDL2-mixer1.2 arm64 1.2.12-19+b1 [81.1 kB]
Get:4 http://deb.debian.org/debian trixie/main arm64 blockout2 arm64 2.5+dfsg1-2 [4,220 kB]
Get:5 http://deb.debian.org/debian trixie/main arm64 cuyo-data all 2.1.0-1 [3,372 kB]
Get:6 http://deb.debian.org/debian trixie/main arm64 cuyo arm64 2.1.0-1+b1 [158 kB]
Fetched 8,041 kB in 1s (12.5 MB/s)
(Reading database ... 198809 files and directories currently installed.)
Removing blockattack (2.9.0-1+b1) ...
Selecting previously unselected package libmikmod3:arm64.
(Reading database ... 198782 files and directories currently installed.)
Preparing to unpack .../0-libmikmod3_3.3.13-1_arm64.deb ...
Unpacking libmikmod3:arm64 (3.3.13-1) ...
Selecting previously unselected package libSDL1.2debian:arm64.
Preparing to unpack .../1-libSDL1.2debian_1.2.68-3_arm64.deb ...
Unpacking libSDL1.2debian:arm64 (1.2.68-3) ...
Selecting previously unselected package libSDL2-mixer1.2:arm64.
Preparing to unpack .../2-libSDL2-mixer1.2_1.2.12-19+b1_arm64.deb ...
Unpacking libSDL2-mixer1.2:arm64 (1.2.12-19+b1) ...
Selecting previously unselected package cuyo-data.
Preparing to unpack .../3-cuyo-data_2.1.0-1_all.deb ...
Unpacking cuyo-data (2.1.0-1) ...
Selecting previously unselected package cuyo.
Preparing to unpack .../4-cuyo_2.1.0-1+b1_arm64.deb ...
Unpacking cuyo (2.1.0-1+b1) ...
Selecting previously unselected package blockout2.
Preparing to unpack .../5-blockout2_2.5+dfsg1-2_arm64.deb ...
Unpacking blockout2 (2.5+dfsg1-2) ...
Setting up blockattack (2.9.0-1+b1) ...
Setting up libmikmod3:arm64 (3.3.13-1) ...
Setting up libSDL1.2debian:arm64 (1.2.68-3) ...
Setting up libSDL2-mixer1.2:arm64 (1.2.12-19+b1) ...
Setting up cuyo-data (2.1.0-1) ...
Setting up cuyo (2.1.0-1+b1) ...
Setting up blockout2 (2.5+dfsg1-2) ...
Processing triggers for libc-bin (2.40-1) ...
```

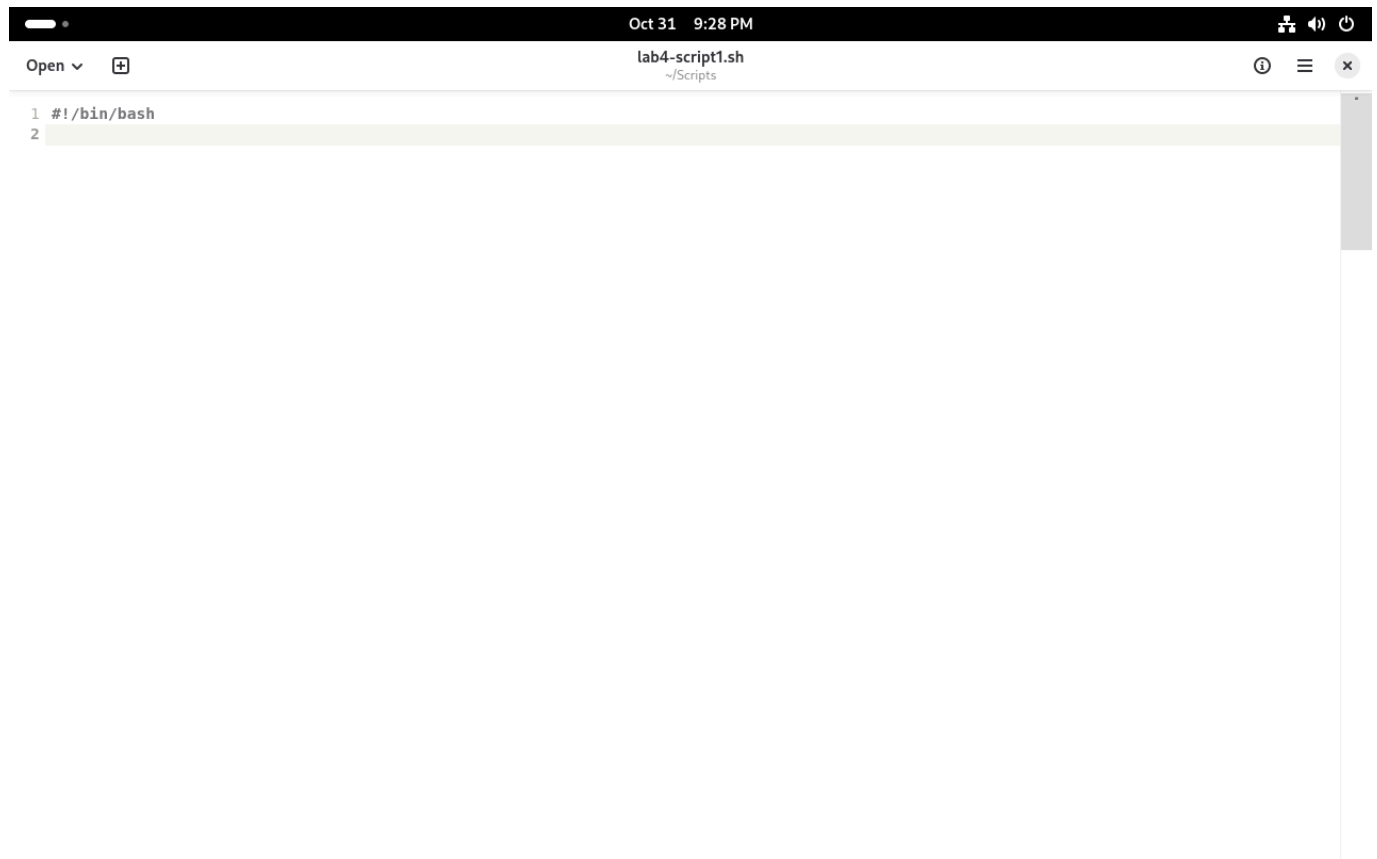


The image shows two screenshots of a terminal window on a system named 'deb@cis106'. The terminal window has a title bar with 'Tilix: deb@cis106: ~' and standard window controls. The first screenshot shows the command 'sudo apt remove cuyo' being executed. The output indicates that several packages were automatically installed and are no longer required: 'cuyo-data', 'libfmt10', 'libopusfile0', 'libphysfs1', 'libsdl2-image-2.0-0', 'libsdl2-mixer-2.0-0', 'libsdl2-ttf-2.0-0', and 'libxmp4'. The terminal shows the removal of 'cuyo' and a summary of the operation, which freed 562 kB of space. The second screenshot shows the command 'sudo apt purge cuyo' being executed. The output indicates that the same set of packages were automatically installed and are no longer required. The terminal shows the removal of 'cuyo*' and a summary of the operation, which freed 7,631 kB of space. Both screenshots show the prompt 'deb@cis106:~\$' and the cursor at the end of the line.

```
1: deb@cis106: ~  
deb@cis106:~$ sudo apt remove cuyo  
The following packages were automatically installed and are no longer required:  
cuyo-data libfmt10 libopusfile0 libphysfs1 libsdl2-image-2.0-0 libsdl2-mixer-2.0-0 libsdl2-ttf-2.0-0 libxmp4  
Use 'sudo apt autoremove' to remove them.  
  
REMOVING:  
cuyo  
  
Summary:  
Upgrading: 0, Installing: 0, Removing: 1, Not Upgrading: 0  
Freed space: 562 kB  
  
Continue? [Y/n] y  
(Reading database ... 199880 files and directories currently installed.)  
Removing cuyo (2.1.0-1+b1) ...  
Processing triggers for hicolor-icon-theme (0.18-2) ...  
Processing triggers for gnome-menus (3.36.0-3) ...  
Processing triggers for mailcap (3.74) ...  
Processing triggers for desktop-file-utils (0.28-1) ...  
deb@cis106:~$ sudo apt purge cuyo  
The following packages were automatically installed and are no longer required:  
cuyo-data libfmt10 libopusfile0 libphysfs1 libsdl2-image-2.0-0 libsdl2-mixer-2.0-0 libsdl2-ttf-2.0-0 libxmp4  
Use 'sudo apt autoremove' to remove them.  
  
REMOVING:  
cuyo*  
  
Summary:  
Upgrading: 0, Installing: 0, Removing: 1, Not Upgrading: 0  
Space needed: 0 B / 26.7 GB available  
  
Continue? [Y/n] y  
(Reading database ... 199871 files and directories currently installed.)  
Purging configuration files for cuyo (2.1.0-1+b1) ...  
deb@cis106:~$  
documentation theme for Jekyll.  
1. Remove the package 'cuyo'
```

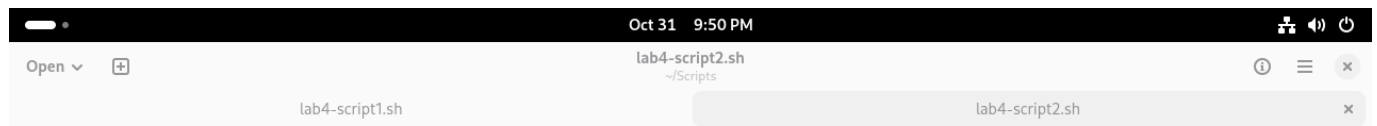
```
1: deb@cis106: ~  
deb@cis106:~$ sudo apt clean && sudo apt autoclean && sudo apt autoremove && sudo apt autopurge  
REMOVING:  
cuyo-data libfmt10 libopusfile0 libphysfs1 libsdl2-image-2.0-0 libsdl2-mixer-2.0-0 libsdl2-ttf-2.0-0 libxmp4  
  
Summary:  
Upgrading: 0, Installing: 0, Removing: 8, Not Upgrading: 0  
Freed space: 7,631 kB  
  
Continue? [Y/n] y  
(Reading database ... 199871 files and directories currently installed.)  
Removing cuyo-data (2.1.0-1) ...  
Removing libfmt10:arm64 (10.1.1+ds1-4) ...  
Removing libsdl2-mixer-2.0-0:arm64 (2.8.1+dfsg-2) ...  
Removing libopusfile0:arm64 (0.12-4+b3) ...  
Removing libphysfs1:arm64 (3.0.2-6+b2) ...  
Removing libsdl2-image-2.0-0:arm64 (2.8.0+dfsg-1) ...  
Removing libsdl2-ttf-2.0-0:arm64 (2.24.0+dfsg-2) ...  
Removing libxmp4:arm64 (4.6.2-2) ...  
Processing triggers for man-db (2.13.1-1) ...  
Processing triggers for libc-bin (2.41-12) ...  
Summary:  
Upgrading: 0, Installing: 0, Removing: 0, Not Upgrading: 0  
deb@cis106:~$  
documentation theme for Jekyll.  
Therefore, to create a shell script, all you need is a text editor. In your virtual machine, you have 3 text editors installed:
```

Question 2



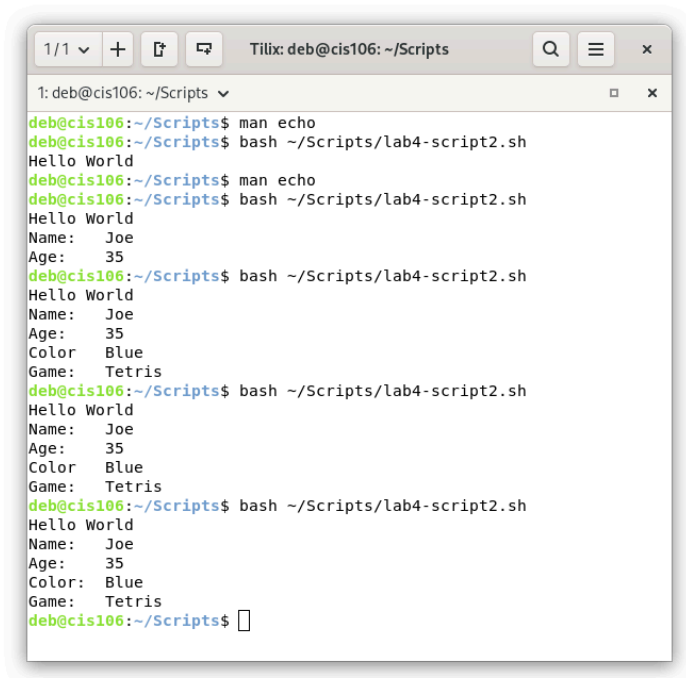
A terminal window titled "lab4-script1.sh" with the path "~/Scripts". The window shows a script with two lines: line 1 is "#!/bin/bash" and line 2 is empty.

```
1 #!/bin/bash
2
```



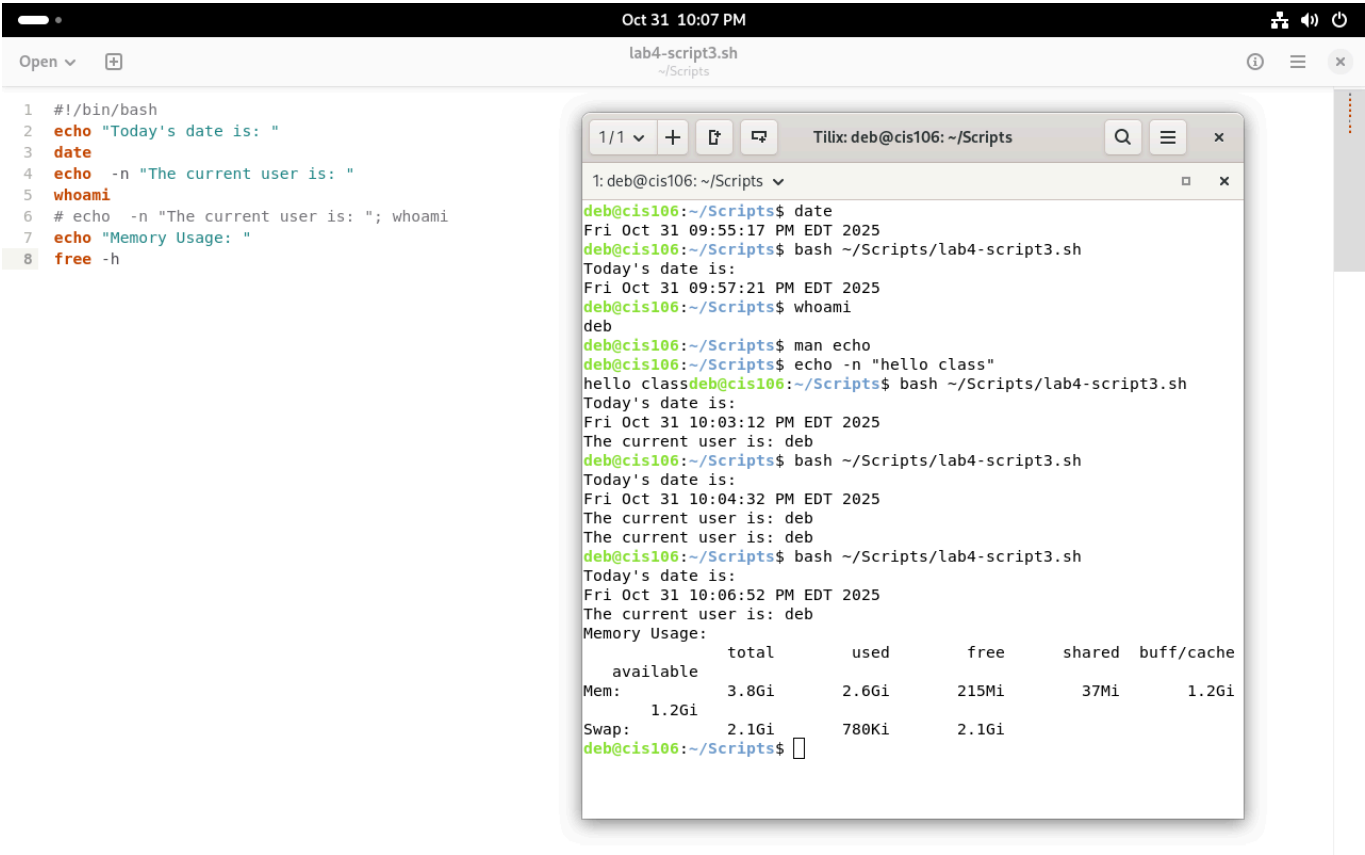
A terminal window titled "lab4-script2.sh" with the path "~/Scripts". The window shows a script with five lines: line 1 is "#!/bin/bash", line 2 is "echo 'Hello World'", line 3 is "echo -e 'Name:\tJoe'", line 4 is "echo -e 'Age:\t35'", and line 5 is "echo -e 'Color:\tBlue\nGame:\tTetris'".

```
1 #!/bin/bash
2 echo "Hello World"
3 echo -e "Name:\tJoe"
4 echo -e "Age:\t35"
5 echo -e "Color:\tBlue\nGame:\tTetris"
```

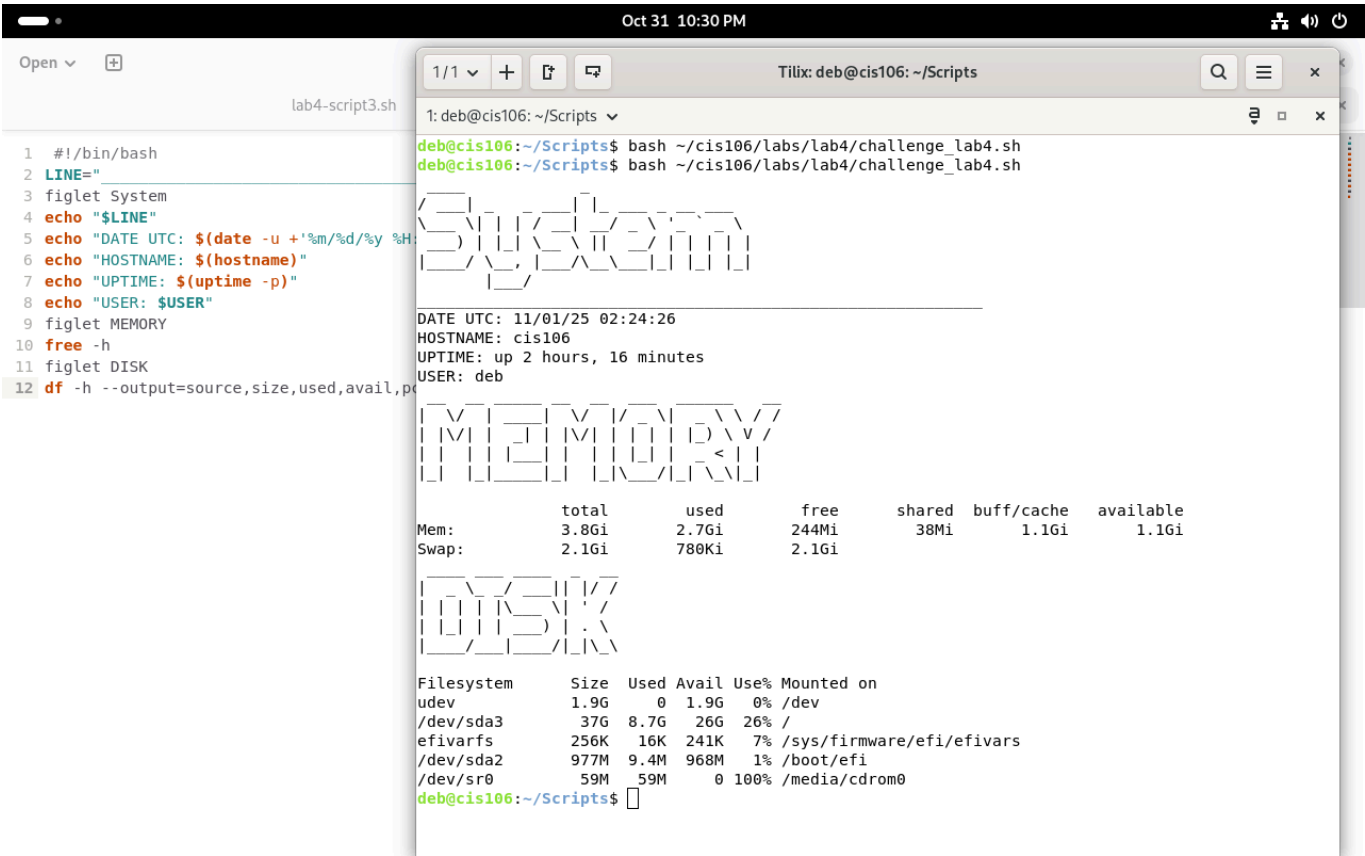


A terminal window titled "Tilix: deb@cis106: ~/Scripts" showing the execution of the script "lab4-script2.sh". The output of the script is displayed, showing "Hello World", "Name: Joe", "Age: 35", "Color: Blue", and "Game: Tetris". The script is executed multiple times, and the output is repeated each time.

```
1/1 v + [ [ [ Tilix: deb@cis106: ~/Scripts [ Q [ x
1: deb@cis106: ~/Scripts v [ x
deb@cis106:~/Scripts$ man echo
deb@cis106:~/Scripts$ bash ~/Scripts/lab4-script2.sh
Hello World
deb@cis106:~/Scripts$ man echo
deb@cis106:~/Scripts$ bash ~/Scripts/lab4-script2.sh
Hello World
Name: Joe
Age: 35
deb@cis106:~/Scripts$ bash ~/Scripts/lab4-script2.sh
Hello World
Name: Joe
Age: 35
Color: Blue
Game: Tetris
deb@cis106:~/Scripts$ bash ~/Scripts/lab4-script2.sh
Hello World
Name: Joe
Age: 35
Color: Blue
Game: Tetris
deb@cis106:~/Scripts$ bash ~/Scripts/lab4-script2.sh
Hello World
Name: Joe
Age: 35
Color: Blue
Game: Tetris
deb@cis106:~/Scripts$
```



Question 3



Scripts